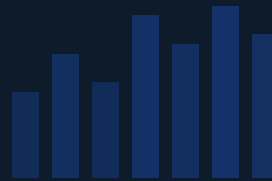


Consumer Behavior Analysis Project

Retail Intelligence • Customer Segmentation • Purchase Driver Analysis



BUSINESS CONTEXT

A leading retail company seeks to deepen its understanding of customer shopping behavior to drive improvements in **sales performance**, **customer satisfaction**, and **long-term loyalty**. Management has observed shifts in purchasing patterns across demographics, product categories, and channels — both online and offline.

KEY PURCHASE DRIVERS

Estimated influence score



"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

FOCUS AREAS



Sales Optimization

Identify which factors drive repeat purchases and revenue.



Customer Loyalty

Segment customers and model long-term retention patterns.



Channel Analysis

Compare online vs. offline behavior and conversion rates.



Marketing Strategy

Personalize campaigns using seasonal and discount data.

PROJECT DELIVERABLES

1

Data Preparation & Modeling (Python)

Clean and transform the raw dataset — handle missing values, encode categories, engineer features for analysis.

2

Data Analysis (SQL)

Organize into structured tables, simulate transactions, and query for insights on segments, loyalty, and purchase drivers.

3

Visualization & Insights (Power BI)

Build an interactive dashboard highlighting key patterns to enable data-driven decisions by stakeholders.

4

Report & Presentation

Write a project report summarizing findings and business recommendations; prepare a visual stakeholder presentation.

5

GitHub Repository

Well-structured repo with all Python scripts, SQL queries, Power BI files, and documentation.

TECHNOLOGY STACK

Tool / Technology	Purpose	Output
Python (Pandas, NumPy)	Data cleaning & feature engineering	Cleaned dataset (.csv)
SQL (PostgreSQL / SQLite)	Structured queries & segmentation	Query results & reports
Power BI	Interactive dashboards	.pbix dashboard file
GitHub	Version control & portfolio sharing	Public repository
PowerPoint / PDF	Stakeholder presentation	Slide deck & report